On the Guidance of Flow Matching: Review on synthetic data

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Motivation & Core Objective

- **Flow Matching (FM):** Generative method learning a velocity field $v_t(x)$ to transport a tractable source distribution p_0 into a complex data distribution p_1.
- **The Question:** How can we generate *specific parts* of the complex target distribution?
- **Solution:** Energy-based guidance! Steer a frozen model toward a defined objective.

1. Notation & Abbreviations Overview

- $p_0 \to p_1$: Source distribution to target data distribution
- $v_t(x)$: Learned base velocity field
- $J(x)$: Energy function (defines the target subset)
- $g_t(x_t)$: Guidance vector field (proportional to $-\nabla J$)
- $v' = v_t + g_t$: Guided velocity field
- $z = (x_0, x_1)$: Coupling of source and target
- $\pi(x_0, x_1)$: Distribution over the source–target coupling
- $\mathcal{P}$: Reverse coupling ratio ($\mathcal{P} = 1$ for independent coupling)

2. Steering Mechanisms

- **Training-time Steering (Conditioning):** Requires data and a label y during training. The model reproduces seen labels directly.
- **Inference-time Steering (Guidance):** Train the model purely on data. Keep the base flow model frozen. Add guidance g_t at sampling to push the flow toward low energy J.

3. Guidance

Guided target distribution. Guidance reweights the original target $p_1(x)$ by the energy J; the subset we want is the low-energy region: $p'(x) = \frac{1}{Z}\, p_1(x)\, e^{-J(x)}$.
Guidance vector field (Theorem 3.1). The field g_t that turns v_t into the guided $v' = v_t + g_t$ is

$$g_t(x_t) = \int \left(\mathcal{P}\, \frac{e^{-J(x_1)}}{Z_t(x_t)} - 1 \right) v_{t|z}(x_t|z)\ \underbrace{p(z|x_t)}_{\text{intractable}}\ dz.$$

The intractable posterior $p(z|x_t)$ is what the methods below approximate. *Training-free* methods spend their cost at inference and trade more assumptions for lower cost; the *training-based* method moves the cost to a one-time training.

- **MC** (g^{MC}) – *training-free, unbiased in the limit.* Monte Carlo estimation of the integral (importance sampling from the prior $p(z)$ instead of the posterior). No gradient of J needed and stays on the data manifold, but has high variance in high dimensions.
- **sim-MC** ($g^{sim\text{-}MC}$) – *training-free.* Approximates the intractable posterior with a Gaussian and samples from it. Useful when J lacks a clean closed-form expression.
- **cov-A** ($g^{cov\text{-}A}$) – *training-free, cheapest.* Taylor-expands the guidance locally around the mean (affine path) and replaces the covariance with a hand-set schedule $\lambda_t I$. No extra network evaluations.
- **cov-G** ($g^{cov\text{-}G}$) – *training-free.* Affine *Gaussian* path: maps FM back to classical diffusion guidance (DPS / ΠGDM) and computes the exact covariance via network Jacobians. Stronger, but costs backprop through the velocity field.
- **GM / Learned** (g_ϕ) – *training-based, exact in expectation.* Guidance Matching: trains one network for the normalizing constant Z and a second that directly learns the guidance field. Accurate, at the price of additional training.

4. The Hidden Assumption ($\mathcal{P} \approx 1$)

- All methods above assume the reverse coupling ratio $\mathcal{P} = \dfrac{\pi'(x_0|x_1)}{\pi(x_0|x_1)} \approx 1$ ($\mathcal{P} = 1$ exactly for independent coupling; $\mathcal{P} \approx 1$ whenever J varies slowly).
- **When it breaks:** g_t points in a systematically wrong direction when coupling is **strong** (e.g. exact Optimal Transport) and the energy J is **sharp**. (Quantified in Section 5.)

5. Results (Synthetic Data)

Setup. A single CFM is trained once on a 2D transport task (e.g. Uniform / 8-Gaussian → 8-Gaussian, 8-Gaussian → Moons), then frozen and steered by an energy J that selects a sub-distribution (here: three of the modes). Guidance methods are compared against unguided CFM and class-conditional CFM (CCFM), which needs the label at training time.

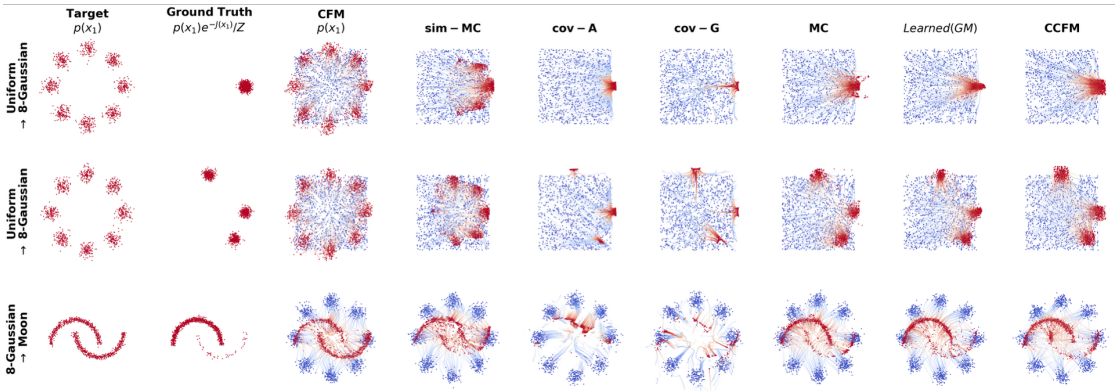


Figure 1: Qualitative comparison. Columns: target p_1, ground-truth guided target $p_1 e^{-J}/Z$, unguided CFM, each guidance method, CCFM (last). Learned (GM) and MC reproduce the selected modes most faithfully; gradient-based approximations (cov-A, cov-G, sim-MC) collapse onto points or drift off the data manifold.

Takeaways. Learned (GM) performs best visually and MC stays on the data manifold. Guidance beats the unguided CFM baseline and matches CCFM *without retraining* – one frozen model generates many sub-distributions, just by changing J. A quantitative comparison over four metrics (W2, energy distance, $|1 -$ diversity trace ratio$|$, off-manifold rate) agrees: Learned (GM) and MC are balanced across all four, while unguided CFM is worst.

Where g^{MC} breaks. The off-manifold rate of g^{MC} (Uniform → 8-Gaussian, energy J for three Gaussians) confirms Section 4: failures grow towards *stronger coupling* (left → right) and *sharper* J (smaller energy width, top of table).

	Coupling strength (weak → strong)					
Energy width	Indep.	Ent. 10	Ent. 1	Ent. 0.3	Ent. 0.1	Exact-OT
0.03 (sharp)	0.57	0.60	0.67	0.74	0.78	0.78
0.05	0.36	0.36	0.47	0.57	0.60	0.62
0.1	0.22	0.18	0.32	0.37	0.41	0.43
0.2 (wide)	0.15	0.10	0.22	0.23	0.29	0.26

Table 1: Off-manifold rate of g^{MC} (lower is better). It climbs towards strong coupling and sharp J.

6. Conclusion

- Introduced different guidance methods for conditional flow matching.
- Learned (GM) performs slightly better than MC on synthetic data, but at the cost of additional training.
- Guidance with g^{MC} under stronger OT coupling and sharp energy functions performs worse.